

# Audio Denoising with Spiking Neural Networks

Dissertation submitted in partial fulfilment of the requirements  
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by

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*Dedicated to my beloved parents*

# Dissertation Approval

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## *Abstract*

Artificial neural networks (ANNs) are the basis of recent advances in artificial intelligence (AI); they typically use real valued neuron responses. In contrast, biological neurons are known to operate using spike trains. In principle, spiking neural networks (SNNs) may have a greater representational capability than ANNs, especially for time series such as speech; we claim to use the representational ability of SNN's for the task of Audio Denoising. Since they are resource efficient, which is due to their sparse and effective representations, we are motivated to incorporate them in place of existing deep learning architectures, which are resource exhaustive. To incorporate these into devices that suppress the noise in an active way, the model should be reasonably lightweight.





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# Abbreviations

|               |                                                |
|---------------|------------------------------------------------|
| <b>STFT</b>   | <b>Short Time Fourier Transform</b>            |
| <b>LPS</b>    | <b>Log Power Spectrum</b>                      |
| <b>NsNet</b>  | <b>Noise supression Net</b>                    |
| <b>GRU</b>    | <b>Gated Recurrent Unit</b>                    |
| <b>LSTM</b>   | <b>Long Short Term Memory</b>                  |
| <b>SI-SDR</b> | <b>Scale Invariant Signal Distortion Ratio</b> |





# Chapter 1

## Introduction

### 1.1 Introduction

Recent years have seen the success of artificial neural networks (ANNs) in speech processing technologies. The neurons traditionally used in these modern networks take real numbers as inputs and produce real-valued outputs. In biological neurons, the information is transmitted in the form of binary sequences of events, called spike trains. The neurons in ANNs can be seen as an instantaneous firing rate approximation of biological spiking neurons, so that the information about the individual timings of the spikes is neglected. Several neuroscience studies suggest that precise spike timings are important in transmitting information, especially in the visual cortex and auditory neurons ([Mainen and Sejnowski, 1995](#)). With the idea of simulating brain-like networks to process information, this firing rate interpretation of spikes can be improved to spiking neuron models. Physiologically plausible mathematical models have been developed to describe neuronal dynamics ([Gerstner and Kistler, 2002](#)). The resulting spiking neurons constitute the building blocks of spiking neural networks (SNNs), and have been called the third generation of neural network models ([Maass, 1997](#)). The temporal dimension of spike trains make them naturally

adapted to sequential input data, such as speech, for which SNNs in principle may have a higher representation capability compared to traditional ANNs (Kasabov, 2019). Another motivation toward SNNs is that the sparsity of spikes over time can allow energy efficient hardware implementations (Davies et al., 2018), (Roy et al., 2019) (Panda et al., 2020) (Dellaferrera et al., 2020), stimulated by event-based sensors, resulting in portable low-powered devices. As pointed out by (Pfeiffer and Pfeil, 2018), this constitutes an advantage over conventional ANNs that rely on energy consuming high-end GPUs. Recent work by (Jeffares et al., 2022) has shown that spike-based techniques can improve not only the ANN efficiency but also its task performance.

## 1.2 Problem Statement

The objective of this project is to develop an effective method for audio denoising, which involves the removal or reduction of unwanted background noise from audio signals while preserving the quality and intelligibility of the original speech or sound

## 1.3 Dissertation Organization

This dissertation is organized into six chapters, including the present introductory chapter (Chapter 1), which outlines the research background, motivation, problem statement, objectives, and a brief overview of the thesis structure. Chapter 2 presents a comprehensive review of the relevant literature, focusing on two key aspects: (i) a comparative analysis between Spiking Neural Networks (SNNs) and traditional Artificial Neural Networks (ANNs) in the context of speech classification, highlighting the advantages of SNNs in terms of data representation efficiency and

reduced model complexity; and (ii) an overview of a deep learning-inspired architecture known as NSNet2. Chapter 3 details the methodology adopted to address the research objectives. Chapter 4 elaborates on the data acquisition process, including data collection and feature extraction procedures. Experimental results and corresponding analyses are presented in Chapter 5. Finally, Chapter 6 summarizes the key findings and outlines potential directions for future work.



# Chapter 2

## Literature Review

### 2.1 Literature Review

In recent years, there has been growing interest in developing biologically inspired neural models for temporal signal processing tasks such as speech recognition and audio analysis. One notable advancement in this area is the work of (Bittar and Garner, 2022), who proposed a novel baseline for speech command recognition using spiking neural networks (SNNs) trained with surrogate gradient. Their work aims to close the performance gap between conventional artificial neural networks (ANNs) and more physiologically plausible spiking models, particularly in time-sensitive auditory applications.

Traditional ANNs, although powerful, process information in a fundamentally different way from biological neurons. They rely on real-valued activations and are generally agnostic to the precise timing of signals. In contrast, SNNs emulate biological neurons by encoding information in spike trains, which are binary and time-discrete

sequences. This approach is thought to offer superior temporal resolution and energy efficiency—two characteristics highly desirable in speech-related applications, especially when deployed on low-power devices.

Despite their potential, SNNs have been historically difficult to train due to the non-differentiable nature of the spiking function, which prevents direct application of gradient-based optimization techniques like backpropagation. To address this, Bittar and Garner utilize surrogate gradient learning, where the non-differentiable spike function is replaced by a smooth approximation (e.g., boxcar or sigmoid functions) during the backward pass. This innovation allows the training of deep SNN architectures within conventional deep learning frameworks like PyTorch, enabling compatibility with widely used toolchains and methodologies in machine learning.

The authors perform an extensive comparison of spiking and non-spiking architectures across several benchmark tasks, including the Spiking Heidelberg Digits (SHD) and Spiking Speech Commands (SSC) datasets. Their experiments involve multiple SNN configurations, including leaky integrate-and-fire (LIF) and adaptive LIF neurons, with and without recurrent connections. Results indicate that adaptive SNNs, when trained using surrogate gradients, not only match but in some cases outperform conventional ANNs, such as GRUs and CNNs, particularly on smaller and more constrained datasets like SHD. For larger tasks like SSC and Google’s Speech Commands (SC), SNNs were able to closely approach ANN performance while maintaining significantly fewer parameters and potentially lower energy consumption.

One of the standout contributions of this work is the demonstration that spike-based architectures can serve as viable and efficient alternatives to traditional models in speech processing tasks. This has direct implications for problems like audio denoising, where temporal precision and energy efficiency are critical. Although

the paper does not directly address denoising, the underlying principles—namely, effective sequential modeling, temporal sparsity, and biologically plausible processing—are highly relevant. Audio denoising, like speech recognition, requires modeling complex temporal dependencies in noisy environments, a task for which SNNs are particularly well-suited due to their intrinsic temporal dynamics and event-driven nature.

Moreover, Bittar and Garner’s work provides a robust training methodology and a comparative framework that can be adapted or extended for other audio tasks, including speech enhancement and noise reduction. Their study lays the groundwork for future exploration into SNN-based models for energy-efficient, real-time audio processing applications.

Now let us jump into the Deep Learning baseline. A significant advancement in the field of speech enhancement and real-time noise suppression was introduced by Microsoft Research through the NSNet2 framework. This model is specifically designed for low-latency, real-time applications, making it highly suitable for deployment on edge devices with constrained computational resources.

NSNet2 builds upon the original NSNet architecture by introducing a more efficient, cascaded structure that decomposes the speech enhancement task into three components: a traditional signal processing front-end, a lightweight neural network, and a refinement stage. This hybrid approach leverages the strengths of both classic DSP techniques and deep learning models to achieve effective noise suppression without incurring the computational costs typical of larger end-to-end neural architectures.

One of the core ideas behind NSNet2 is to pre-enhance the noisy input using a spectral subtraction-based DSP method, which produces a coarse but clean signal estimate. This estimate is then fed, along with the original noisy input, into a

compact DNN, which learns a mapping to further refine the enhancement quality. Finally, a post-processing block ensures artifact reduction and temporal consistency. This design allows NSNet2 to perform well even on low-power CPUs and embedded systems, making it highly relevant for on-device real-time applications such as video conferencing, telephony, and assistive hearing devices.

In terms of performance, NSNet2 strikes a balance between accuracy, latency, and computational cost. It achieves notable improvements in PESQ, STOI, and SNR metrics, outperforming baseline DSP-only methods while being significantly faster than more complex DNN-based approaches. The system is capable of processing 10 ms audio frames with less than 10 ms total latency, ensuring seamless user experiences in real-time settings.

For research focused on audio denoising, NSNet2 is particularly valuable due to:

Its edge-friendliness, which aligns with the increasing demand for offline, private audio processing.

Its robustness across diverse noise conditions, which is critical for practical denoising in unconstrained environments.

Its modular design, which offers potential for further enhancement using advanced components such as spiking neural networks, attention-based layers, or spectral masking techniques. ([Braun and Tashev, 2020](#))

## 2.2 Summary

Recent advancements in biologically inspired and low-latency neural architectures have opened promising directions for audio denoising and speech enhancement tasks.



Bittar and Garner (2022) proposed a spiking neural network (SNN) baseline for speech command recognition that leverages surrogate gradient training to overcome the challenges of non-differentiable spike functions. Their models—based on leaky integrate-and-fire (LIF) and adaptive LIF neurons—demonstrate competitive or superior performance to traditional artificial neural networks (ANNs) on benchmark tasks, all while maintaining high temporal resolution and energy efficiency, particularly well-suited for temporal auditory data.

In parallel, Dubey et al. (2020) introduced NSNet2, a real-time noise suppression framework designed for deployment on edge devices. It combines classic signal processing with lightweight neural network modules to create a hybrid system that offers strong noise suppression with very low latency and minimal computational overhead. NSNet2 emphasizes robustness, modularity, and practicality, delivering high PESQ/STOI performance on-device without requiring heavy deep learning pipelines.

While the SNN approach offers a biologically plausible, event-driven model potentially useful for future ultra-low-power and temporally-aware applications, NSNet2 represents a mature, deployable solution optimized for immediate real-world use. Together, these works illustrate complementary pathways toward effective, efficient audio enhancement—one grounded in neuromorphic inspiration, and the other in engineering pragmatism and deployment-readiness.



# Chapter 3

## Approach and Methods used

### 3.1 Introduction

Initially, the baseline model, NSNet2, was implemented and trained on a benchmark audio dataset to assess its performance. The core components of NSNet2's architecture are Gated Recurrent Units (GRUs), which are designed to capture the sequential dependencies inherent in audio signals. To explore the potential benefits of biologically inspired computation, the GRU blocks were replaced with various types of spiking neurons. The resulting architectures were then evaluated and compared in terms of parameter count, performance metrics, and computational efficiency. Subsequently, different hyperparameters of the modified models were systematically varied to further investigate their impact on overall performance.

## 3.2 Models and Algorithms

### 3.2.1 NsNet2

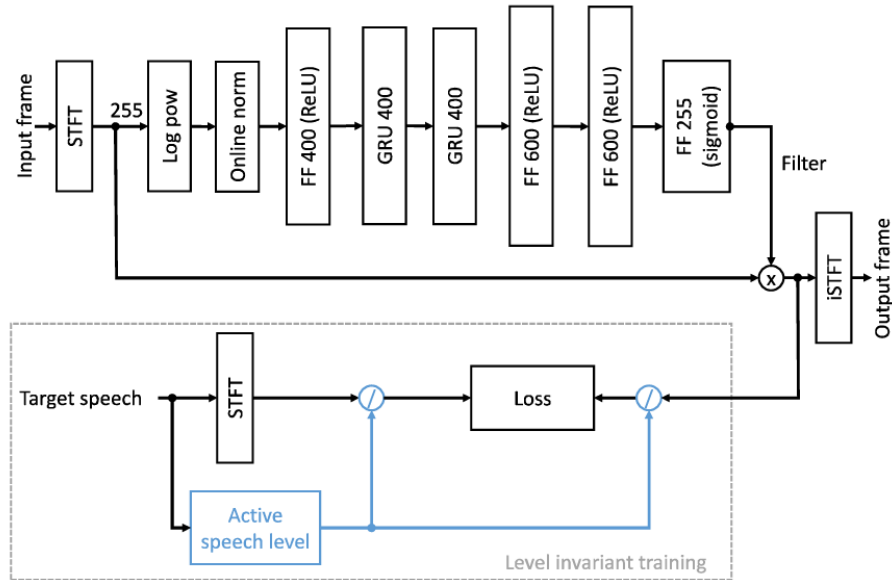


FIGURE 3.1: The NsNet2 architecture

The NsNet2 model is a lightweight and efficient deep learning architecture designed specifically for real-time speech enhancement tasks, particularly single-channel noise suppression. It builds upon the foundation of the original NsNet architecture, introducing several key improvements to enhance performance while maintaining low computational complexity suitable for deployment on edge devices.

The overall architecture consists of three primary components:

#### 1. Spectral Feature Extraction:

The input noisy speech waveform is first transformed into a spectral representation using a Short-Time Fourier Transform (STFT). The log-power spectrum (LPS) of the noisy input serves as the primary input feature to the model. This

transformation effectively captures the frequency-domain characteristics of the signal, which are critical for distinguishing speech from noise.

## **2. Recurrent Modeling with GRUs:**

The core of NsNet2’s architecture is a stack of Gated Recurrent Unit (GRU) layers. These recurrent layers are responsible for modeling the temporal dependencies across speech frames. GRUs are preferred for their ability to capture long-term dependencies while being computationally more efficient than Long Short-Term Memory (LSTM) networks. NsNet2 uses a multi-layer GRU design, where each layer processes the sequential output from the previous one, allowing for hierarchical temporal feature extraction.

## **3. Spectral Mask Estimation and Reconstruction:**

The output from the final GRU layer is passed through a feedforward layer to estimate a spectral mask, which is used to suppress noise components in the input signal. This mask is applied (element-wise multiplication) to the complex spectrum of the noisy speech. The enhanced time-domain waveform is reconstructed using the inverse STFT (iSTFT).

Importantly, NsNet2 achieves a favorable trade-off between accuracy and computational efficiency, making it suitable for on-device inference. The model has been optimized for low-latency applications, with significant reductions in parameter count and memory usage compared to more complex deep learning-based speech enhancement models.

### **3.2.2 Spiking Neuron Models**

This section outlines the spiking neuron models evaluated in this study, which serve as fundamental computational units in Spiking Neural Networks (SNNs) for speech

command recognition. These models are inspired by the biophysical properties of real neurons and aim to balance biological realism with computational efficiency.

### 3.2.2.1 Leaky Integrate-and-Fire (LIF) Neuron

The Leaky Integrate-and-Fire (LIF) neuron is one of the simplest and most widely adopted spiking neuron models. It describes the membrane potential  $u(t)$  of a neuron, which evolves over time according to the following differential equation:

$$\tau_u \frac{du(t)}{dt} = -(u(t) - u_{\text{rest}}) + RI(t) \quad (3.1)$$

Here,  $\tau_u$  is the membrane time constant,  $u_{\text{rest}}$  is the resting potential,  $R$  is the membrane resistance, and  $I(t)$  is the input current. When the membrane potential reaches a threshold  $\vartheta$ , the neuron emits a spike and the potential is reset to a value  $u_r$ .

### 3.2.2.2 Recurrent Leaky Integrate-and-Fire (RLIF) Neuron

The Recurrent Leaky Integrate-and-Fire (RLIF) neuron extends the standard LIF model by incorporating recurrent synaptic connections within each layer. This allows neurons to integrate not only feedforward inputs from the previous layer but also feedback from other neurons in the same layer at previous time steps. The total input current for a neuron at time  $t$  becomes:

$$I_i^l[t] = \sum_{j=1}^{N^{l-1}} W_{ji}^l s_j^{l-1}[t] + \sum_{k=1, k \neq i}^{N^l} V_{ki}^l s_k^l[t-1] + b_i^l \quad (3.2)$$

where  $W^l$  and  $V^l$  are the feedforward and recurrent weight matrices, respectively, and  $b_i^l$  is a bias term. The remaining membrane potential dynamics follow the standard discrete LIF formulation as defined earlier.

### 3.2.2.3 Adaptive Leaky Integrate-and-Fire (adLIF) Neuron

The Adaptive LIF (adLIF) neuron extends the basic LIF model by introducing an adaptation current  $w(t)$ , which allows the neuron to exhibit firing patterns such as spike-frequency adaptation. The dynamics are governed by:

$$\tau_u \frac{du(t)}{dt} = -(u(t) - u_{\text{rest}}) - Rw(t) + RI(t) \quad (3.3)$$

$$\tau_w \frac{dw(t)}{dt} = -w(t) + a(u(t) - u_{\text{rest}}) + \tau_w b \sum_f \delta(t - t^f) \quad (3.4)$$

where  $\tau_w$  is the adaptation time constant,  $a$  controls subthreshold adaptation, and  $b$  controls spike-triggered adaptation via a delta function at firing times  $t^f$ .

### 3.2.2.4 Recurrent Adaptive Leaky Integrate-and-Fire (RadLIF) Neuron

The Recurrent Adaptive Leaky Integrate-and-Fire (RadLIF) neuron combines the adaptation dynamics of the adLIF neuron with the recurrent connectivity of the RLIF model. It maintains the discrete-time equations for membrane potential and adaptation current:

$$u[t] = \alpha u[t-1] + (1 - \alpha)(I[t] - w[t-1]) - \vartheta s[t-1] \quad (3.5)$$

$$w[t] = \beta w[t-1] + (1 - \beta)au[t-1] + bs[t-1] \quad (3.6)$$

$$s[t] = \mathbb{I}(u[t] \geq \vartheta) \quad (3.7)$$

with the input current  $I[t]$  including both feedforward and recurrent components as in Equation (10). This model captures both temporal dependencies and adaptive firing behaviors, making it the most expressive and biologically inspired model considered in this study.

### 3.2.2.5 Discrete-Time Implementation

To facilitate implementation in modern deep learning frameworks, the continuous-time adLIF model is discretized using Euler integration with a step size  $\Delta t = 1$  ms. The resulting equations are:

$$u[t] = \alpha u[t-1] + (1 - \alpha)(I[t] - w[t-1]) - \vartheta s[t-1] \quad (3.8)$$

$$w[t] = \beta w[t-1] + (1 - \beta)au[t-1] + bs[t-1] \quad (3.9)$$

$$s[t] = \mathbb{I}(u[t] \geq \vartheta) \quad (3.10)$$

where  $s[t] \in \{0, 1\}$  is the binary spike output, and  $\alpha = \exp(-\Delta t/\tau_u)$ ,  $\beta = \exp(-\Delta t/\tau_w)$ .

These spiking neuron models form the computational basis for the SNN architectures evaluated in this work. Among them, the adLIF model was found to provide the best trade-off between biological plausibility and task performance.



## 3.3 Methods and Experiments

### 3.3.1 Experiment 1: Replacement of GRU Blocks with Spiking Neuron Models

In this experiment, we aimed to evaluate the effectiveness of spiking neural networks (SNNs) as a replacement for the Gated Recurrent Unit (GRU) blocks in the original NsNet2 architecture. To this end, we systematically substituted the GRU-based recurrent layers with SNN blocks constructed using four different types of spiking neuron models: Leaky Integrate-and-Fire (LIF), Adaptive LIF (adLIF), Recurrent LIF (RLIF), and Recurrent Adaptive LIF (RadLIF). These neuron models were selected based on their increasing levels of biological realism and computational complexity.

Across all configurations, replacing the GRU blocks with spiking neuron blocks led to a substantial reduction in the total number of trainable parameters. Specifically, while the baseline NsNet2 model comprised approximately 2.78 million parameters, the hybrid architectures with spiking neuron models contained only around 1.13 million parameters, reflecting a reduction of approximately 60% in model size. This demonstrates the efficiency of SNN-based designs in terms of memory and computational requirements.

Among the four variants, the architecture employing adLIF neurons achieved the most favorable performance after 11 training epochs. The inclusion of adaptive mechanisms in the adLIF model enabled it to better capture temporal dependencies in speech signals, allowing it to closely match, and in some instances compete with, the performance of the baseline GRU-based NsNet2 model. This finding highlights

the potential of adaptive spiking neurons in designing efficient yet expressive models for real-time speech enhancement.

To assess model performance, we employed the Scale-Invariant Signal-to-Distortion Ratio (SI-SDR), a widely accepted objective metric in the fields of speech enhancement and source separation. Unlike conventional Signal-to-Distortion Ratio (SDR), SI-SDR eliminates the influence of global scale differences between the clean reference and the estimated signal, thereby providing a more perceptually meaningful evaluation.

Given a clean target signal  $s$  and an estimated signal  $\hat{s}$ , SI-SDR is computed by projecting  $\hat{s}$  onto  $s$  to determine the scale-aligned target component:

$$s_{\text{target}} = \frac{\langle \hat{s}, s \rangle}{\|s\|^2} s \quad (3.11)$$

The residual error component is then defined as:

$$e_{\text{noise}} = \hat{s} - s_{\text{target}} \quad (3.12)$$

Finally, the SI-SDR is given by:

$$\text{SI-SDR} = 10 \log_{10} \left( \frac{\|s_{\text{target}}\|^2}{\|e_{\text{noise}}\|^2} \right) \text{ dB} \quad (3.13)$$

A higher SI-SDR score indicates a lower level of distortion and better reconstruction quality. Due to its robustness, scale-invariance, and interpretability, SI-SDR has become a standard metric for evaluating speech enhancement models in recent literature.

### 3.3.2 Experiment 2: Effect of Surrogate Gradient Functions

In this experiment, we investigated the impact of different surrogate gradient functions on the performance of spiking neural networks when integrated into the hybrid architecture. Surrogate gradient functions are essential in training spiking neurons using backpropagation, as the true gradient of the spiking activation function is non-differentiable. To approximate the gradient during training, we experimented with three commonly used surrogate functions:

- **Gaussian** function
- **Exponential** function
- **Multi-Gaussian** function

In each case, the surrogate gradient was applied to approximate the derivative of the spiking nonlinearity. That is, the gradient of the spike function was replaced by the chosen surrogate during the backward pass. The models were trained for 100 epochs under each surrogate function to ensure a fair and consistent comparison. In between, we changed certain parameters of the functions to improve the training.

It is important to note that in the previous experiment (Experiment 1), a *box-car* surrogate gradient function was used as the default. The box-car function is defined to have a constant gradient of 1 within a fixed interval (typically  $[-0.5, 0.5]$ ) and zero elsewhere, providing a simple rectangular approximation of the spike derivative.

After training, we evaluated the models using the SI-SDR metric. Interestingly, the results across all surrogate gradient functions were observed to be similar, with no significant variation in SI-SDR scores. This suggests that, for the task of speech

enhancement in this hybrid architecture, the choice among these surrogate gradients does not critically impact overall model performance.

### **3.3.3 Experiment 3: Effect of Threshold Values on Model Performance**

In this experiment, we explored the impact of varying neuronal threshold values on the performance of the spiking neuron-based architecture. The threshold determines the membrane potential level at which a neuron emits a spike, and thus plays a critical role in regulating spiking activity and information flow through the network. While a threshold value of 1 was used as the default in previous experiments, we extended the investigation by experimenting with three distinct threshold values: 0.05, 5, and 10. These values were deliberately chosen to span a wide dynamic range in order to examine whether extreme or subtle thresholds could lead to observable changes in network behavior or performance.

The model was trained with each of the threshold settings while keeping all other hyperparameters constant. Performance was again evaluated using the SI-SDR metric. Similar to the results observed with different surrogate gradient functions, variations in the threshold value did not lead to any significant differences in the final performance. This suggests that the model is relatively robust to the choice of spiking threshold within the tested range, indicating that precise tuning of this parameter may not be critical for achieving effective speech enhancement.

### 3.4 Summary

This study explored a biologically inspired modification of the NsNet2 architecture for real-time speech enhancement by integrating spiking neural network (SNN) components. The baseline NsNet2 model employs a spectral feature extractor, stacked Gated Recurrent Unit (GRU) layers for temporal modeling, and a spectral mask estimation module to reconstruct clean speech from noisy input. To evaluate the potential of SNNs in this setting, the GRU layers were replaced with spiking neuron blocks using various neuron models: Leaky Integrate-and-Fire (LIF), Adaptive LIF (adLIF), Recurrent LIF (RLIF), and Recurrent Adaptive LIF (RadLIF).

In **Experiment 1**, this substitution led to a substantial reduction in model complexity, with the number of trainable parameters decreasing by approximately 60%. Among the variants, the model utilizing adLIF neurons demonstrated the best performance after 11 training epochs, effectively competing with the original GRU-based model in terms of SI-SDR, a standard metric for evaluating speech enhancement quality.

In **Experiment 2**, the effect of different surrogate gradient functions—Gaussian, Exponential, and Multi-Gaussian—was investigated. These surrogate functions were used to approximate the non-differentiable spike function during backpropagation. Despite varying the form and parameters of these functions, the results showed negligible differences in SI-SDR, suggesting that the hybrid architecture is largely insensitive to the specific choice of surrogate gradient.

In **Experiment 3**, we analyzed the impact of varying spiking thresholds on model behavior, testing values of 0.05, 5, and 10, with 1 as the default. The results again indicated minimal variation in SI-SDR across these settings, implying that the model is robust to threshold variations within this range.

Overall, the experimental findings highlight the efficiency and flexibility of integrating spiking neuron models into the NsNet2 framework. The use of adaptive spiking neurons (e.g., adLIF) offers a favorable balance between performance and parameter efficiency, while the model remains stable across variations in surrogate gradients and threshold values.



# Chapter 4

## Dataset

### 4.1 Introduction

For the training and evaluation of the speech enhancement models, we utilized the publicly available VCTK-DEMAND noisy speech corpus, which is hosted on the University of Edinburgh’s DataShare platform. This dataset is constructed by artificially contaminating clean speech recordings from the Voice Cloning Toolkit (VCTK) corpus with real-world noise samples from the Diverse Environments Multichannel Acoustic Noise Database (DEMAND).

The corpus provides parallel pairs of clean and noisy utterances, making it particularly well-suited for supervised learning tasks in speech enhancement and denoising. The speech recordings feature multiple native English speakers with varying accents, and are originally sampled at 48 kHz.

The dataset is organized into training and test partitions, with two primary configurations: a 28-speaker set and a 56-speaker set. In this study, we employed the



28-speaker subset, which contains approximately 10,000 utterances for training and 800 utterances for testing. Each noisy utterance is generated by mixing a clean speech sample with one of ten noise types at various signal-to-noise ratios (SNRs), ensuring a diverse and realistic testing environment.

The availability of clean-noisy pairs, along with the inclusion of a wide range of noise types and speaker variability, makes the VCTK-DEMAND dataset a widely adopted benchmark in the field of speech enhancement.

## 4.2 Detailed Description of the Dataset

The VCTK-DEMAND noisy speech corpus is a synthetic dataset designed for training and evaluating speech enhancement algorithms in realistic acoustic conditions. It combines clean speech recordings from the Voice Cloning Toolkit (VCTK) corpus with background noise samples from the DEMAND (Diverse Environments Multi-channel Acoustic Noise Database) to create parallel pairs of clean and noisy speech signals. This structure supports supervised learning methods and facilitates objective evaluation of model performance.

### 4.2.1 Speech Data (VCTK Corpus)

The clean speech component of the dataset is derived from the VCTK corpus, which contains recordings from 109 native English speakers with various accents. Each speaker reads a set of approximately 400 phonetically balanced sentences, sampled at 48 kHz in a semi-anechoic chamber using a high-quality microphone setup. For the noisy speech corpus, a subset of these speakers is used:

- **Training Set:** 28 speakers
- **Test Set:** 2 speakers (different from the training set to evaluate generalization)

This speaker diversity introduces accent and articulation variability, making the dataset representative of real-world speech characteristics.

### 4.2.2 Noise Data (DEMAND Corpus)

To generate noisy utterances, each clean waveform is mixed with one of ten environmental noise recordings from the DEMAND database. These noise types are drawn from various acoustic environments such as domestic, office, public transport, and street settings. Importantly, the noises are real recordings (not synthetically generated), contributing to the realism of the resulting noisy speech.

The noise is added at three signal-to-noise ratio (SNR) levels: **0 dB, 5 dB, and 10 dB**, simulating challenging noise conditions. Each noisy utterance is paired with its clean counterpart, enabling frame-level supervised learning.

### 4.2.3 Sampling Rate and Preprocessing

All audio samples in the dataset are provided at a sampling rate of 48 kHz. During preprocessing, we perform the following steps:

1. Trimming or zero-padding waveforms to a fixed length
2. Normalization of amplitude values

#### 4.2.4 Data Partitioning and Volume

The 28-speaker training subset consists of approximately 10,000 utterances, each paired with three noisy versions (corresponding to different SNRs and noise types), resulting in a total of roughly 30,000 training samples. The test set includes around 824 utterances from two unseen speakers, mixed with the same noise types at the same SNR levels.

#### 4.2.5 Use in Literature

The VCTK-DEMAND dataset is widely used as a benchmark in speech enhancement literature, especially in deep learning-based models such as SEGAN, DCCRN, MetricGAN, and BSRNN. Its structured clean-noisy alignment, speaker variability, and realistic noise conditions make it ideal for evaluating generalization, robustness, and real-time inference capability.

#### 4.2.6 Relevance to This Work

In this study, we use the VCTK-DEMAND dataset to train and evaluate both the baseline NsNet2 model and its hybrid variants incorporating spiking neuron models. The availability of parallel clean and noisy speech pairs, along with the dataset's diversity, allows for robust training and fair evaluation using objective metrics such as SI-SDR and SNR.

### 4.2.7 Data Extraction

Each audio waveform was transformed into the frequency domain using the Short-Time Fourier Transform (STFT). This operation converts the time-domain signal into a time-frequency representation, which is more suitable for tasks such as speech enhancement. For each waveform, the log power spectrogram (LPS) was computed, providing a perceptually relevant and numerically stable feature for model training.

The STFT was implemented using an FFT size (`n_fft`) of 508. The hop length and window length were both set to one-fourth of the FFT size, i.e.,

$$\text{hop\_length} = \frac{\text{n\_fft}}{4} = 127, \quad \text{win\_length} = \frac{\text{n\_fft}}{4} = 127$$

This configuration ensures sufficient overlap between successive frames while balancing time and frequency resolution. The resulting log power spectrograms were then used as input features for both the baseline and spiking neural network-based models.



# Chapter 5

## Results and Discussions

### 5.1 Experiment 1 results:

TABLE 5.1: Performance comparison of different spiking neuron models used in the NsNet2 architecture.

| Neuron | SI-SDR (dB) | SNR (dB) | Epoch | Trainable Parameters | Param. Reduction |
|--------|-------------|----------|-------|----------------------|------------------|
| adLIF  | 14.04       | 10.96    | 11    | 1,132,663 (1.13M)    | 60%              |
| LIF    | 8.53        | 4.98     | 11    | 1,129,927 (1.13M)    | 60%              |
| RLIF   | 11.86       | 11.39    | 11    | 1,275,463 (1.27M)    | 55%              |
| RadLIF | 13.97       | 10.74    | 11    | 1,278,199 (1.28M)    | 55%              |

It is observed that the hybrid model incorporating adLIF neurons consistently outperformed the other spiking variants. In particular, the adLIF-based architecture achieved significantly better performance compared to models utilizing LIF and RLIF neurons. Furthermore, it also demonstrated little better results relative to the RadLIF-based model. This performance gap may be attributed to potential overfitting in the more complex RadLIF architecture, suggesting that the adaptive dynamics of adLIF neurons strike a more effective balance between model complexity and generalization for the given task.

## 5.2 Experiment 2 results:

As said earlier, we tried different surrogate functions. Here are the results.

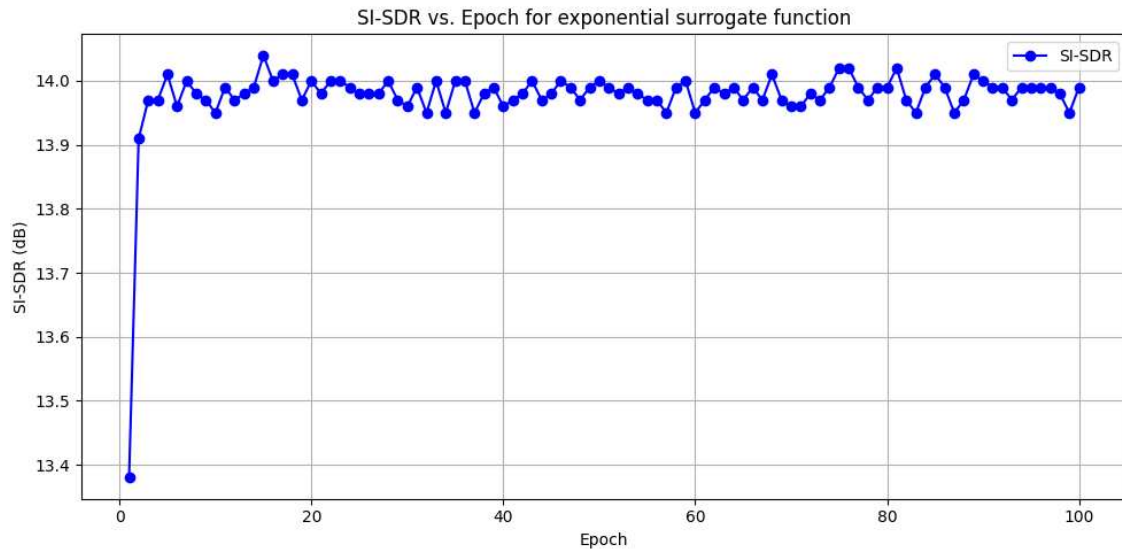


FIGURE 5.1: SI-SDR versus Epoch plot for exponential surrogate function

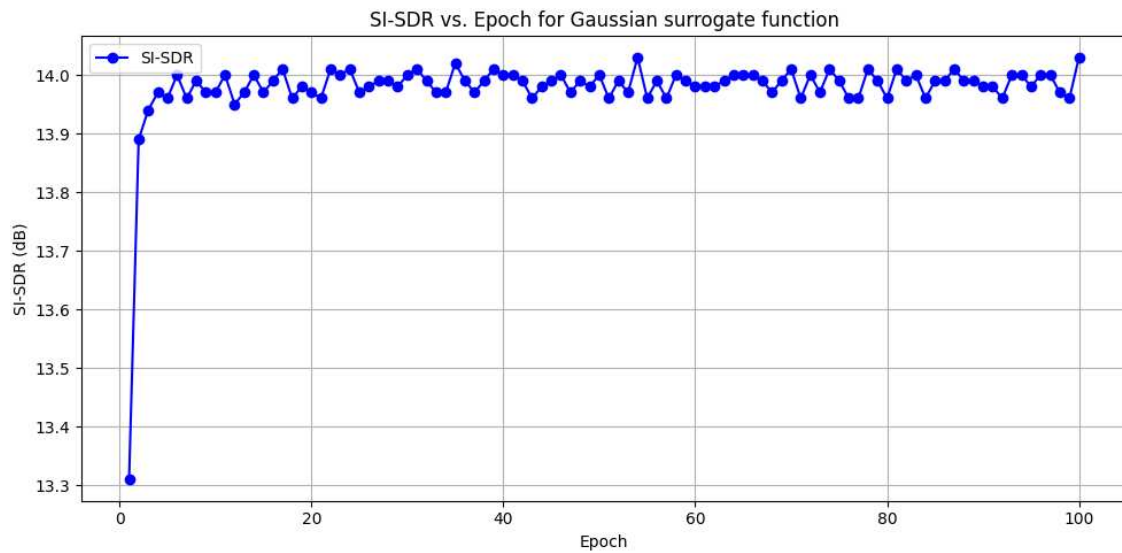


FIGURE 5.2: SI-SDR versus Epoch plot for Gaussian surrogate function

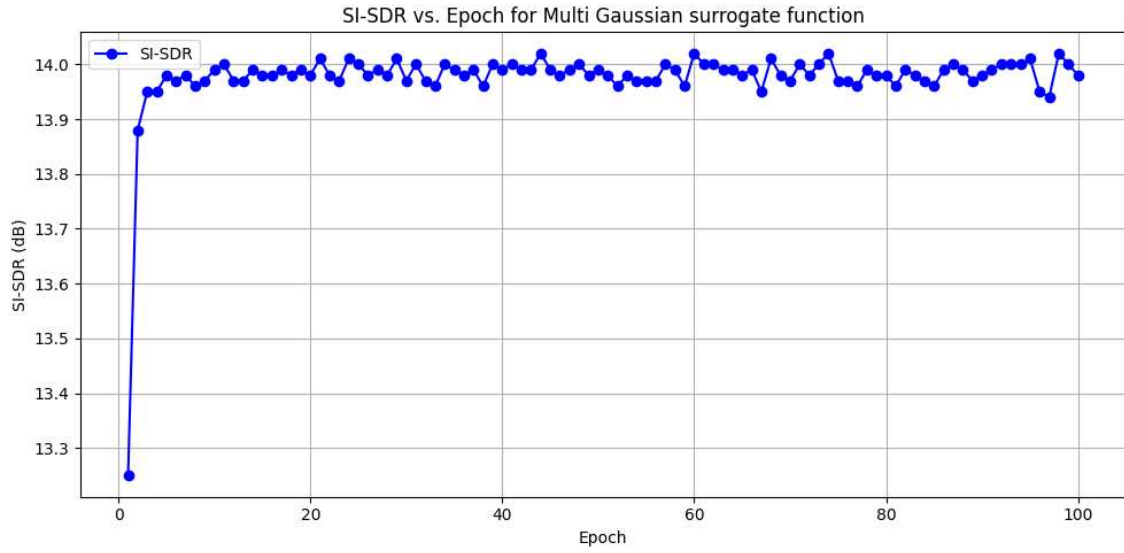


FIGURE 5.3: SI-SDR versus Epoch plot for Multi Gaussian surrogate function

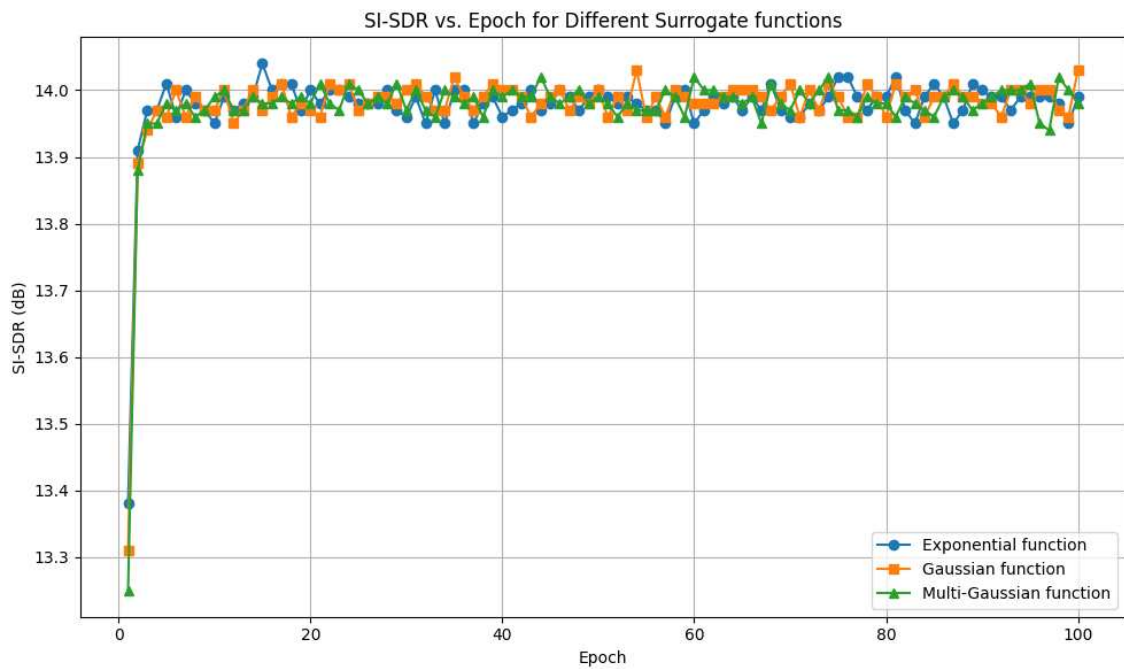


FIGURE 5.4: SI-SDR versus Epoch plot for different surrogate functions



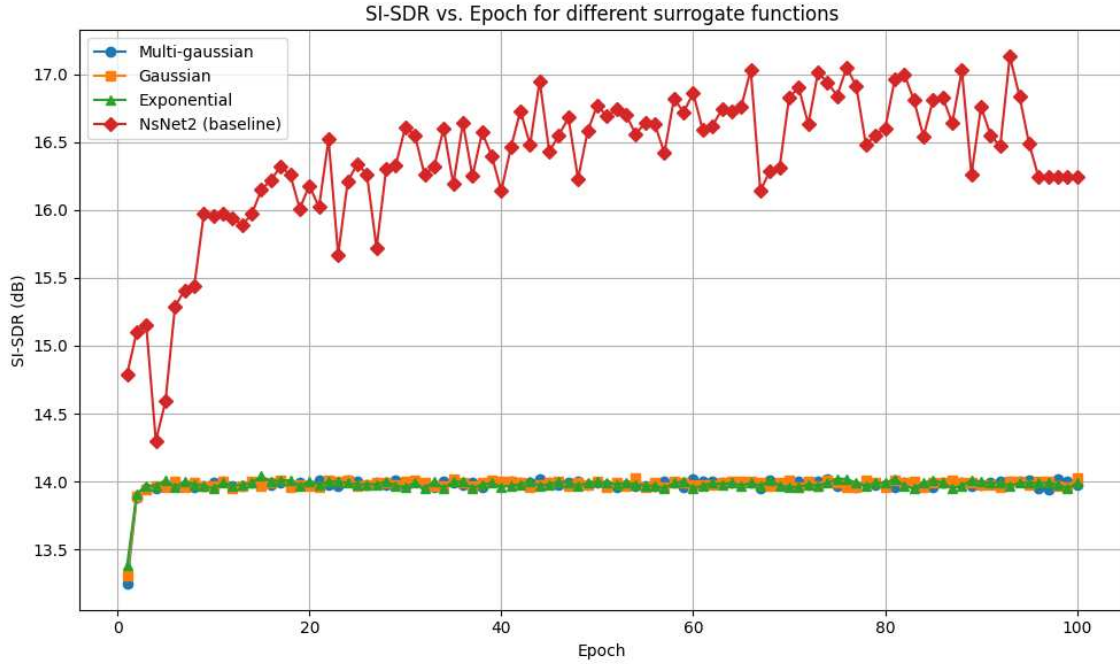


FIGURE 5.5: SI-SDR versus Epoch plot for different surrogate functions with baseline

Note: The baseline has been trained for 96 epochs (the last 4 values are padded with the SI-SDR of 96th epoch)

We observe that that the baseline has performed better than the hybrid models. We observe that across all surrogate gradient functions employed in this study, there is no significant variation in the resulting performance metrics, such as SI-SDR. This consistency in performance suggests that, for the current dataset and task of speech enhancement, the specific choice of surrogate function does not play a critical role in influencing the model's effectiveness. All tested surrogate functions yielded comparable results across multiple trials and experimental configurations.

This outcome implies a degree of robustness in the training dynamics of spiking neural networks with respect to the surrogate gradient formulation. It is possible that the dominant factors affecting model performance lie in architectural choices

(e.g., neuron type, recurrent connections, adaptation mechanisms) or dataset characteristics, rather than in the exact functional form of the surrogate gradient.

Overall, the minimal impact of the surrogate function observed in this experiment simplifies the model design process, as it allows researchers to select computationally efficient or easily implementable surrogate functions without compromising model performance. Further investigation on more complex or diverse datasets could help establish whether this insensitivity generalizes beyond the current experimental setting.

### 5.3 Experiment 3 results:

As said earlier, we tried different threshold values. Here are the results.

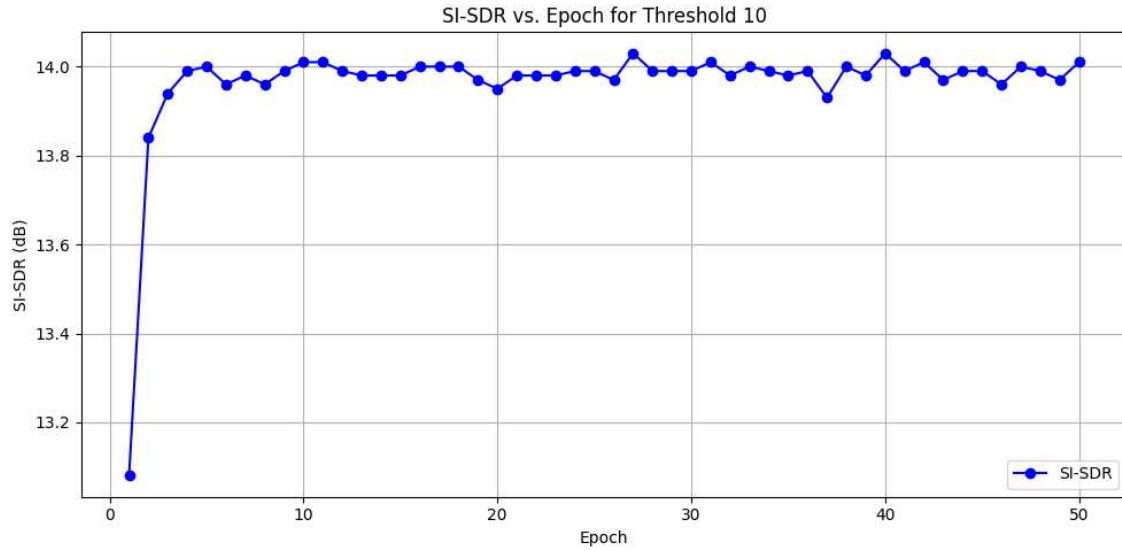


FIGURE 5.6: SI-SDR versus Epoch plot for Threshold 10

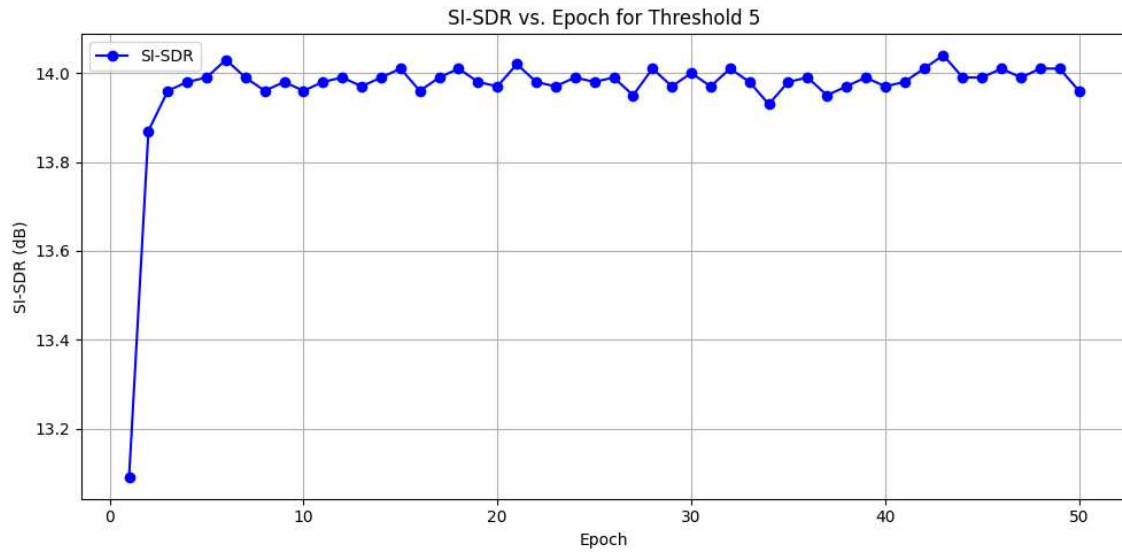


FIGURE 5.7: SI-SDR versus Epoch plot for Threshold 5

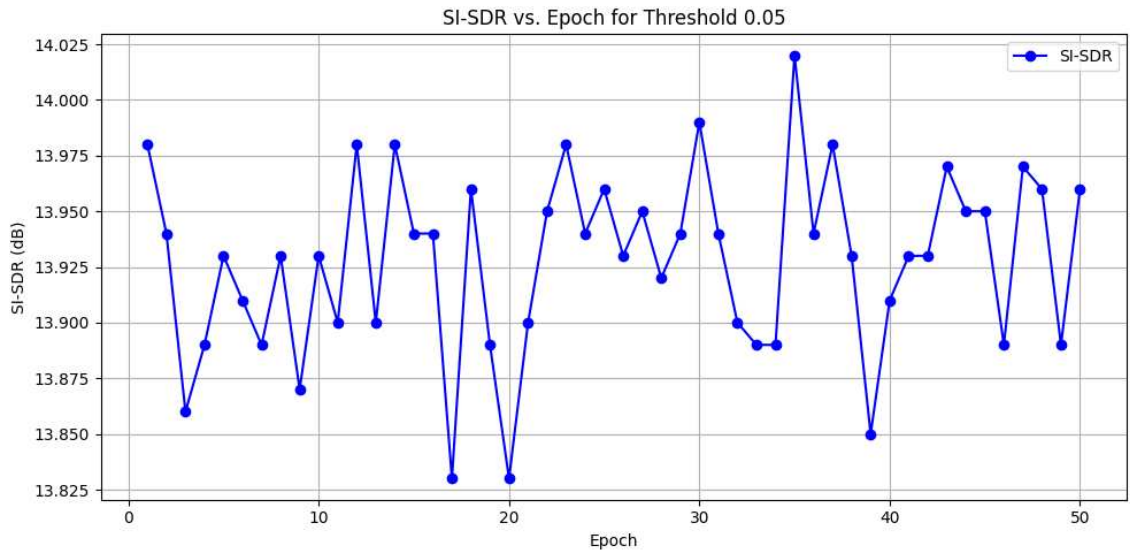


FIGURE 5.8: SI-SDR versus Epoch plot for Threshold 0.05

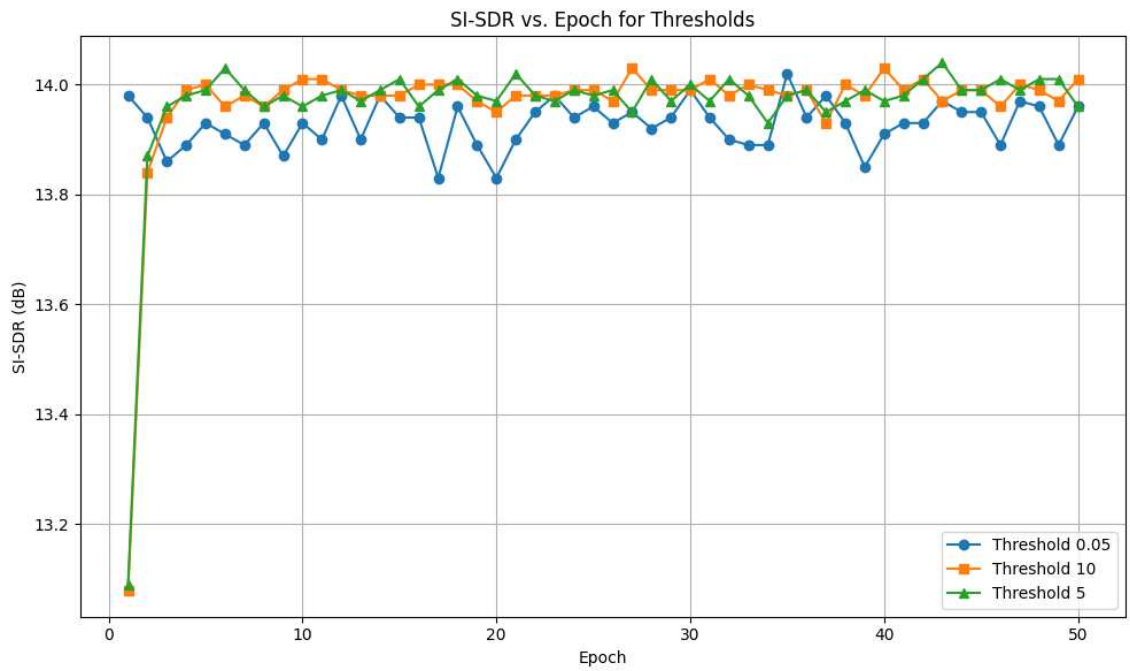


FIGURE 5.9: SI-SDR versus Epoch plot for different thresholds

We observe that varying the neuronal threshold values does not lead to significantly different results across the evaluated performance metrics. This suggests that, within the tested range of thresholds, the spiking neural network maintains a degree of robustness with respect to this parameter. The relative insensitivity of the model to changes in threshold implies that fine-tuning this hyperparameter may not be critical for achieving optimal performance on the current dataset. It is possible that other architectural components or hyperparameters, such as adaptation dynamics or learning rates, exert a more dominant influence on model behavior. Further experiments with a wider threshold range or under different noise conditions could provide deeper insight into its impact.



# Chapter 6

## Summary and Conclusions

### 6.1 Conclusions

In this study, we proposed a hybrid neural architecture that replaces the GRU blocks in the NsNet2 speech enhancement model with biologically inspired spiking neuron models. The key finding is that the hybrid models are able to achieve performance comparable to the original baseline model while significantly reducing the number of trainable parameters—by approximately 60%. Although the hybrid models did not outperform the baseline in terms of SI-SDR, their competitive performance despite substantial parameter reduction demonstrates the potential of spiking neural networks (SNNs) for efficient, real-time speech processing.

Among the four spiking neuron models investigated which are :Leaky Integrate-and-Fire (LIF), Adaptive LIF (adLIF), Recurrent LIF (RLIF), and Recurrent Adaptive LIF (RadLIF), the adLIF neuron consistently achieved the best performance. This

suggests that the inclusion of adaptation mechanisms enhances the temporal modeling capabilities of the network, making adLIF neurons particularly suitable for speech enhancement tasks.

An important observation from additional experiments is that the choice of surrogate gradient function and the spiking threshold value had minimal impact on overall model performance. This indicates that the architecture is relatively robust to these hyperparameters, simplifying the design and tuning process.

## 6.2 Future Work

For future work, we propose incorporating attention mechanisms to further improve the model's performance and capacity to capture long-range dependencies. Additionally, exploring more complex spiking neuron models such as the Adaptive Exponential Integrate-and-Fire (AdEx) and Izhikevich neurons may offer further insights into the trade-offs between biological realism and computational efficiency in speech enhancement systems.





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